

Hopfield Network Exercise

Neural Computing

February 25, 2026

1 Pattern Storage & Weight Matrix Calculation

Consider a Hopfield network with 9 neurons arranged in a 3×3 grid. We will use bipolar representation where each neuron can either be +1 or -1.

We want to store the following two patterns:

- Pattern A: A horizontal line

$$\begin{bmatrix} +1 & +1 & +1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

- Pattern B: A vertical line

$$\begin{bmatrix} +1 & -1 & -1 \\ +1 & -1 & -1 \\ +1 & -1 & -1 \end{bmatrix}$$

Questions:

1. Convert patterns A and B into vectors $\vec{\xi}^A$ and $\vec{\xi}^B$, where each vector has 9 elements.
2. Using the Hebbian learning rule, calculate the weight matrix W . Recall that:
 - $w_{ij} = \frac{1}{N} \sum (\xi_i^\mu \xi_j^\mu)$ for all patterns μ .
 - The diagonal elements w_{ii} should be set to 0 (no self connections).
 - Show your calculations for at least 3 specific weights.

2 Pattern Retrieval

3. Consider a new input pattern C

$$\begin{bmatrix} +1 & +1 & +1 \\ -1 & -1 & -1 \\ +1 & -1 & -1 \end{bmatrix}$$

This pattern has one neuron different from Pattern A. Now:

- (a) Convert pattern C to vector form $\vec{\xi}$.
- (b) Calculate the state of each neuron after one synchronous update. Recall that the update rule is:

$$O_i(t+1) = \text{sgn} \left(\sum_j^N w_{ij} O_j(t) \right)$$

- (c) Does the network converge to one of the stored patterns after this update? If not, continue the updates until convergence or until you can explain why it won't converge.

3 Energy Function

The energy function of a Hopfield network is given by:

$$E(\vec{x}) = -\frac{1}{2} \sum_i^N \sum_j^N w_{ij} \xi_i \xi_j$$

Questions:

4. Calculate the energy of Pattern A ($\vec{\xi}^A$), Pattern B ($\vec{\xi}^B$), Pattern C ($\vec{\xi}^C$).
5. Based on your energy calculations, explain:
 - (a) Why do the stored patterns have lower energy values?
 - (b) How the network uses energy minimization to perform pattern completion?